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APPENDIX: PROJECT ELECTRONIC FOLDER STRUCTURE

1. PURPOSE

This procedure purpose is to describe **IUI's Document Management System use for XXXXX Project.**

Especially the following points:

- **Project electronic folder management.**
- **Every project role access to the information.** IUI's staff, partners, suppliers and client staff
- **Documents import and management.**

2. INTRODUCTION

IUI owns technology to integrate the whole company's information flow. This technology is managed by **Documentum 4i, a powerful Document Manager** with the following properties and functionalities:

1) **Intranet/Internet oriented system:** Using a standard browser (Internet Explorer, i.e.) you can create, modify or browse any project document. Using a web environment ease remote information access.
2) **Indexed information:** for every document there are properties (document id., title, description…) that provide a direct link to documents and search services and tools.
3) **Information easy access:** Advanced search using documents attributes. Through the folder structure you can link documents making them available from different locations (the same document)
4) **Information security and Integrity:** automatic security procedures based on access list checking (ACL: Access Control List) for documents and folders defining every user groups privileges. (read, write, delete...). This system guarantees that every user accesses allowed information and provides error protection.
5) **Version management:** when you modify any file (document) you can overwrite (loosing the original NOT APPROVED VERSION if you are allowed) or keep the older version in the documents base. This option provides you an accessible document life cycle history.
6) **Electronic messenger:** Improves users' communication attaching documents to the message (usually the attached document is the message itself)

7) **Lifecycles and workflows management.** Lifecycle is a document attribute to describe document state; it's always attached to the document and any state of document lifecycle (creation, approval, customer revision) is reflected in an attribute change.

   These state changes are automated in **workflows**. You can program the flow of approval: a sequence of users (recorded in a register) approve the document until its final approval.

8) **Multi-format purpose:** You can import documents in their original format (AutoCAD .dgn, excel .xls, word .doc....) and it's available to include, attached to the original or by its own, an standard electronic document publisher (Acrobat .pdf)

3. DOCUMENTS INCLUDED IN THE PROJECT

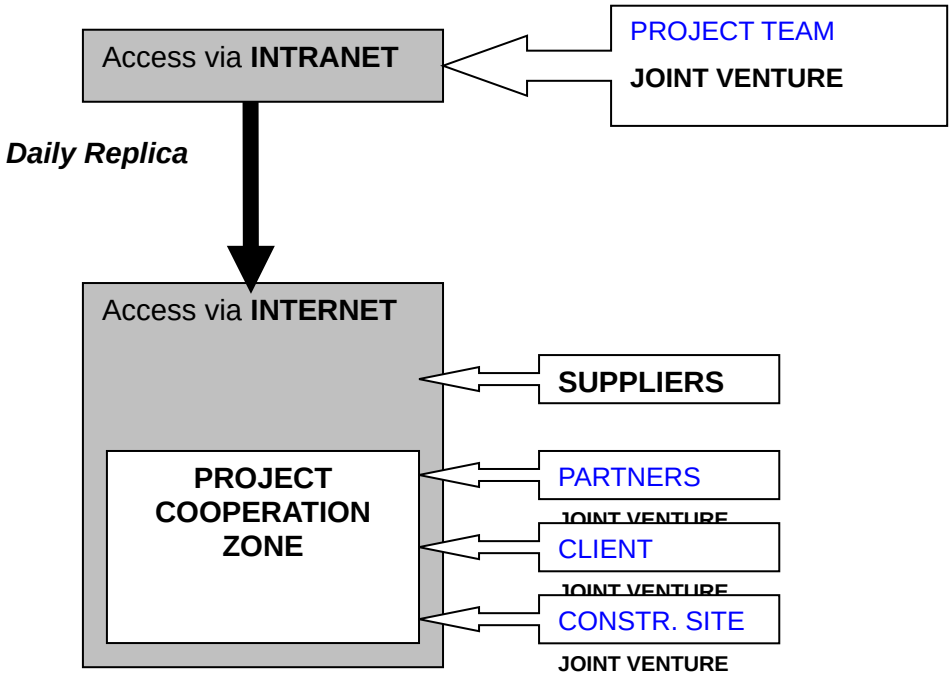
All of the project documents will be imported into the Document Base:

- **Customer documents:**
  - **Technical documents.** Basic Engineering, specifications and standards, previous installation drawings…
  - **Purchasing and subcontract documents:** requests for quotation, commercial evaluations, orders…
- **EPCM Contractor documents:**
  - **Detail Engineering documents**: drawings, specifications, data sheets…
  - **Requisition of Material** and **Requisition of Construction (Material Supply** and **Subcontracts).**
  - **General documents:** Progress reports, meetings, quality program
- **Suppliers' documents.**

All of the project documents will be imported to the Document Base by **IUI's staff** and will be available for the whole company with the permissions developed, avoiding not allowed access to the information in the Intranet.

The main point in e-business is security; we have two Document Bases isolated by a firewall: one for intranet and the other for Internet. Everyday we make a copy of the intranet content to the Internet Document Base so we can provide remote access to the information to our customers, suppliers, partners...

4. ACCESS STRUCTURE



4.1 CLIENT ACCESS

We provide two customer accesses, one for resident customer staff and the other for remote access using Internet:

- **Resident customer staff** may access via Intranet to project documents. They will also be involved in the electronic approval flow when requested.
- **Internet access.** Remote information access is available through Intecsa Uhde web site (**http://www.intecsauhde.com/web/english/root.htm**) entering **Project Cooperation Zone (http://194.179.91.198/rs-bin/RightSite.dll/ssi4.htm)**

  This link allows access to all of the **approved** project documents (or the ones that approval is not required). Access will not be allowed during the approval process (workflow) of current edition, only previously approved versions will be available.

4.2 PARTNERS ACCESS

- Partners may access to the whole project documents with the access level agreed in the contract. Remote access is available through Intecsa Uhde web site (**http://www.intecsauhde.com/web/english/root.htm**) entering **Project Cooperation Zone (http://194.179.91.198/rs-bin/RightSite.dll/ssi4.htm)**

4.3. SUPPLIERS ACCESS

Using **Business Center Access** in Intecsa Uhde web site suppliers will find all of the **Material Supply documents** related to them either in Inquiry phase or Order phase. Business Center information **security is guaranteed by an access code list (user name and password)** for the suppliers managed by IUI Procurament Department so just the request for quotation invited suppliers or the order selected ones may view the information

5. DOCUMENTS IMPORT, APPROVAL AND DISTRIBUTION

Every document **created by Intecsa Uhde** will be imported to the Document Base by document's creator and it would be electronically approved and distributed if necessary.

The Project Administrator who will be also responsible for the project electronic cabinet management will import customer documents.

The Project Administrator will also be the one responsible for the **supplier documents** management, importing them into the Document Base when they arrive at Intecsa Uhde, and when they are sent back to the suppliers with comments.

5.1. Engineering Documents

As we said before **Project Administrator** will manage the project electronic cabinet, in especial he/she is the person in charge of:

- **Electronic Documents distribution.**
- **Customer documents importation to the Document Base**

- **Material Supply documents attachment** to ease suppliers' access to the information.
- **Project Electronic Cabinet management:** error detection, documents properties and permissions supervising.
- **Suppliers' documents dispatch record.**
- When it's needed to work with **heavy lots of documents** (importing or extracting from the Document Base) he/she will be able to use **specific tools (see XXXXX-IT-03 "Electronic File Exchange Procedure").**

5.2. Suppliers documents

Project Administrator is in charge of:

- **Supplier expediting**
- **Document import to the Document Base. (Arrival)**
- **Make documents available for the engineers in charge of the order.**
- **Import documents once IUI staff had commented them,** so documents and their comments are available for the project staff, client and the involved supplier.

6. DOCUMENTATION STRUCTURE

The structure design is oriented to:

- **Quick information access.**
- **Improve permission management and ease documents modifications.**

Indexing criteria is oriented firstly to sort documents into different Project areas, then activities and later into document type (customer documents, mailing...).

The RM´s (Relation of Material supplies) are developed in a matrix structure, this means that each department that issues RM´s has got several foulders for each type of document and one specific foulder named "RM´s & Related Documents" which contains three groups of documents: one foulder with links to all engineering documents listed in the List of Attached Documents(LDA) of the RM, another foulder with the Request for Quotation documents and a third foulder with vendor Documents.

Documents are generated and imported into their technical speciality folder, for example, a pump data sheet will be located at:

06.1. MECHANICAL – ROTATING EQUIPMENT

3.0 DATA SHEETS

- And if the document is included in the Relation of Material supply document, it will be also **linked (not copied)** to that material supply document folder:

6.0. RELATIONS OF MATERIAL and RELATED DOCUMENTS

RM-XXXX- .....

RM DOCUMENTS. RM – XXXX-.....

See APENDIX 1 for the complete project structure and their different access levels to the information contained.

7. PROJECT USERS AND PERMISSIONS.

Documentum (Doc. Base) users are organized in user groups so permissions **(ACL. Access Control Lists)** are given to these groups. To guarantee information security you just need to include agents in their appropriate group.

For any modification in project team, permissions, or for updating (include new agents or delete old ones) **Project Management** must contact **IUI´s Knowledge Management Department.**

ACLs are structured by disciplines.

In principle, technical documents are accessible to all Project staff. Other documents regarding economic and comercial matters will be assigned restricted permissions so that they will be accessed only by authorised personnel.

Project Administrator will have full access to all Project Documents.

The ACL´s system is flexible enough to be adapted to Project needs.

Approved documents by Workflow are protected against overwriting or erasure.

AVAILABLE PROJECT PERMISSIONS

HOLD

Customer permissions have been described in the Client Access point

APPENDIX:

PROJECT ELECTRONIC FOLDER STRUCTURE

Here we have the complete project folders structure and we also describe with the following codes their content access permissions

1) **ctm:** client users.
2) **pm:** project management
3) **cost**: project cost control leader.
4) **dle:** discipline leader engineers
5) **dm:** department managers
6) **iui:** Intecsa Uhde staff

For example,

pm + cost + dm means Project Management, Project Cost Control Leader, Department Managers are allowed to access to folder content.

Other clarifications**:**

- It refers to general-purpose folders**.**
- It refers to **Relation of Material Supplies** or **Subcontracts** folders**.**

PROJECT'S FOLDER STRUCTURE	ACCESS PERMISSIONS
XXXXX- INDORAMA PROJECT	
01. INFORMATION FROM CLIENT	
AWARD PACKAGE	
DESIGN INFORMATION	
SPECIFICATIONS AND STANDARS	
INSULATION	
PAINTING	
CIVIL AND ARCHITECTURAL	
PIPING	
ROTATING EQUIPMENT	
INSTRUMENTATION	
ANALYTICAL INSTRUMENTS	
ELECTRICAL	
MATERIALS	
PROCESS CONTAINMENT	
02. GENERAL	
PERMITTING	
CIVIL AND STRUCTURAL PROJECT	
MECHANICAL AND INSTALLATION PROJECTS	
_ELECTRICAL INSTALLATION PROJECT	
PROJECT MANAGEMENT AND CONTROL	
1.0. COORDINATION MANUAL	
2.0. PROJECT INSTRUCTIONS	

PROJECT'S FOLDER STRUCTURE	ACCESS PERMISSIONS
3.0 MINUTES OF MEETINGS	
4.0. SCHEDULES	
5.0. PROGRESS REPORTS	
6.0. CHANGE ORDERS	
03. BASIC ENGINEERING	
04. CIVIL AND ARCHITECTURE	
1.0. CALCULATIONS	
2.0. TECHNICAL BID ANALYSIS	
3.0. DESIGN CRITERIA	
4.0. SPECIFICATIONS	
5.0. MEASURINGS AND BOQ's	
6.0. DRAWINGS	
05. ELECTRICAL	
_ELECTRICAL INSTALLATION PROJECT	
1.0. CALCULATIONS	
2.0. TECHNICAL BID ANALYSIS	
3.0. SPECIFICATIONS and STANDARDS	
4.0. DATA SHEETS	
4.1. LISTS and MEASURINGS	
5.0. DRAWINGS	
6.0. REQUISITIONS OFMATERIAL SUPPLIES (RM)	
7.0. REQUISITIONS OF MATERIAL and RELATED DOCUMENTS	
RM-.....	
RMs DOCUMENTS : RM-….	

PROJECT'S FOLDER STRUCTURE	ACCESS PERMISSIONS
REQUEST FOR QUOTATION DOCUMENTS : RM-….	
VENDOR DOCUMENTS: RM-....	
07. INSTRUMENTATION	
1.0. CALCULATIONS	
2.0. TECHNICAL BID ANALYSIS	
3.0. SPECIFICATIONS and STANDARDS	
4.0. DATA SHEETS	
4.1. LISTS and MEASURINGS	
5.0. DRAWINGS and DIAGRAMS	
ASSEMBLY DIAGRAMS	
CONTROL LOGIC DIAGRAMS	
CONTROL LOOP DIAGRAMS	
JUNCTION BOXES and INTERCONEXIONS	
6.0. REQUISITIONS OFMATERIAL SUPPLIES (RM)	
7.0. REQUISITIONS OFMATERIAL and RELATED DOCUMENTS	
07.1. MECHANICAL – ROTATING EQUIPMENT	
1.0. CALCULATIONS	
2.0. TECHNICAL BID ANALYSIS	
2.0. SPECIFICATIONS	
3.0. DATA SHEETS	
4.0. DRAWINGS and SKETCHES	
5.0. REQUISITIONS OFMATERIAL SUPPLIES (RM)	
6.0. REQUISITION OFMATERIAL SUPPLIES and RELATED DOCUMENTS	

PROJECT'S FOLDER STRUCTURE	ACCESS PERMISSIONS
07.2. MECHANICAL – STATIC EQUIPMENT	
1.0. CALCULATIONS	
2.0. TECHNICAL BID ANALYSIS	
3.0. SPECIFICATIONS and STANDARDS	
4.0. DATA SHEETS	
5.0. DRAWINGS	
6.0. REQUISITIONS OFMATERIAL SUPPLIES (RM)	
7.0. REQUISITIONS OFMATERIAL SUPPLIES and RELATED DOCUMENTS	
07.3. MECHANICAL – PIPING MATERIALS	
1.0. CALCULATIONS	
2.0. TECHNICAL BID ANALYSIS	
3.0. SPECIFICATIONS and STANDARDS	
4.0. DATA SHEETS	
5.0. LISTS	
6.0. REQUISITIONS OF MATERIAL SUPPLIES (RM)	
7.0. REQUISITIONS OFMATERIAL SUPPLIES and RELATED DOCUMENTS	
07.4. MECHANICAL – STRESS & PIPING SUPPORTS	
1.0. CALCULATIONS	
2.0. TECHNICAL BID ANALYSIS	
3.0. SPECIFICATIONS	
4.0. DATA SHEETS	
5.0. DRAWINGS	
6.0. REQUISITIONS OFMATERIAL SUPPLIES (RM)	

PROJECT'S FOLDER STRUCTURE	ACCESS PERMISSIONS
7.0. REQUISITIONS OFMATERIAL SUPPLIES and RELATED DOCUMENTS	
07.5. MECHANICAL – FIRE PROTECTION	
1.0. CALCULATIONS	
2.0. TECHNICAL BID ANALYSIS	
3.0. SPECIFICATIONS	
4.0. MEASURINGS	
5.0. DRAWINGS	
6.0. REQUISITIONS OF MATERIAL SUPPLIES (RM)	
7.0. REQUISITIONS OF MATERIAL SUPPLIES and RELATED DOCUMENTS	
08. PIPING DESIGN	
1.0. STANDARDS	
2.0. ISOMETRIC DRAWINGS	
3.0. TIE-IN's	
4.0. PLOT PLANS	
5.0. LAY-OUT DRAWINGS	
6.0. RACK DRAWINGS	
09. PROCESS	
1.0. CALCULATIONS	
2.0. DESIGN BASIS, DESIGN CRITERIA AND SPECIFICATIONS	
3.0. DATA SHEETS	
EQUIPMENT	
INSTRUMENTS	
4.0. LISTS	

PROJECT'S FOLDER STRUCTURE	ACCESS PERMISSIONS
5.0. DRAWINGS	
PROCESS FLOW DIAGRAM (PFDs)	
PIPING & INSTRUMENT DIAGRAMS (P&IDs)	
PLOT PLANS	
10. PROCUREMENT	
1.0. GENERAL	
2.0. COMERCIAL EVALUATIONS	
3.0.ORDERS	
4.0. INSPECTION	
11. CONSTRUCTION	
1.0. SPECIFICATIONS	
2.0. COMERCIAL EVALUATIONS	
3.0. ORDERS	
4.0. SUBCONTRACTS (RC)	
5.0. SUBCONTRACTS and RELATED DOCUMENTS	
RC-.....	
RCs DOCUMENTS : RC :….	
REQUEST FOR QUOTATION DOCUMENTS : RC-…	
SUPPLIERS DOCUMENTS: RC-...	